

NAME : SHAH FAISAL

ROLL NO : 23I-0058

SECTION : BSE – 4A

OPERATING SYSTEMS LAB 11

BSSE SPRING 2026

LAB 11

TASK 01 :

```
#include <stdio.h>
#include <pthread.h>
int num1, num2;
double result_add, result_sub, result_mul, result_div;
void* addition(void* arg) {
    result_add = num1 + num2;
    printf("Addition:    %d + %d = %.2f\n", num1, num2, result_add);
    return NULL;
}
void* subtraction(void* arg) {
    result_sub = num1 - num2;
    printf("Subtraction:  %d - %d = %.2f\n", num1, num2, result_sub);
    return NULL;
}
void* multiplication(void* arg) {
    result_mul = num1 * num2;
    printf("Multiplication: %d * %d = %.2f\n", num1, num2, result_mul);
    return NULL;
}
void* division(void* arg) {
    if (num2 != 0)
        result_div = (double)num1 / num2;
    else
        result_div = 0;
    printf("Division:    %d / %d = %.2f\n", num1, num2, result_div);
    return NULL;
}
int main() {
    pthread_t t_add, t_sub, t_mul, t_div;
    printf("=== Multithreaded Calculator ===\n");
    printf("Enter num1: ");
    scanf("%d", &num1);
    printf("Enter num2: ");
    scanf("%d", &num2);
    printf("\n");
    pthread_create(&t_add, NULL, addition,    NULL);
    pthread_create(&t_sub, NULL, subtraction,  NULL);
```

```
pthread_create(&t_mul, NULL, multiplication, NULL);
pthread_create(&t_div, NULL, division, NULL);
pthread_join(t_add, NULL);
pthread_join(t_sub, NULL);
pthread_join(t_mul, NULL);
pthread_join(t_div, NULL);
printf("\n--- Final Results ---\n");
printf("Addition Result: %.2f\n", result_add);
printf("Subtraction Result: %.2f\n", result_sub);
printf("Multiplication Result: %.2f\n", result_mul);
printf("Division Result: %.2f\n", result_div);

return 0;
}
```

```
shah@DESKTOP-02Q2MPC:~$ gcc lab11_task1.c -o task1
shah@DESKTOP-02Q2MPC:~$ ./task1
=== Multithreaded Calculator ===
Enter num1: 12
Enter num2: 13

Addition:      12 + 13 = 25.00
Subtraction:   12 - 13 = -1.00
Multiplication: 12 * 13 = 156.00
Division:      12 / 13 = 0.92

--- Final Results ---
Addition Result:      25.00
Subtraction Result:   -1.00
Multiplication Result: 156.00
Division Result:      0.92
```

TASK 02:

```
#include <stdio.h>
#include <pthread.h>
#include <math.h>
typedef struct {
    double input;
    double result;
} ThreadData;
```

```
void* compute_square(void* arg) {
    ThreadData* data = (ThreadData*)arg;
    data->result = data->input * data->input;
    printf("Square:    %.2f * %.2f = %.2f\n",
        data->input, data->input, data->result);
    return NULL;
}

void* compute_sqrt(void* arg) {
    ThreadData* data = (ThreadData*)arg;
    if (data->input < 0) {
        printf("Square Root: Cannot compute sqrt of negative number %.2f\n", data->input);
        data->result = 0;
    } else {
        data->result = sqrt(data->input);
        printf("Square Root: sqrt(%.2f) = %.4f\n",
            data->input, data->result);
    }
    return NULL;
}

int main() {
    double number;
    printf("=== Multithreaded Calculator (Square & Square Root) ===\n");
    printf("Enter a number: ");
    scanf("%lf", &number);
    ThreadData sq_data = {number, 0};
    ThreadData sqrt_data = {number, 0};
    pthread_t t_square, t_sqrt;
    pthread_create(&t_square, NULL, compute_square, (void*)&sq_data);
    pthread_create(&t_sqrt, NULL, compute_sqrt, (void*)&sqrt_data);
    pthread_join(t_square, NULL);
    pthread_join(t_sqrt, NULL);
    printf("\n--- Final Results ---\n");
    printf("Square of %.2f:    %.2f\n", number, sq_data.result);
    printf("Square Root of %.2f: %.4f\n", number, sqrt_data.result);
    return 0;
}
```

```
shah@DESKTOP-02Q2MPC:~$ gcc lab11_task2.c -o task2 -lpthread -lm
shah@DESKTOP-02Q2MPC:~$ ./task2
=== Multithreaded Calculator (Square & Square Root) ===
Enter a number: 12
Square:      12.00 * 12.00 = 144.00
Square Root: sqrt(12.00) = 3.4641

--- Final Results ---
Square of 12.00:      144.00
Square Root of 12.00: 3.4641
shah@DESKTOP-02Q2MPC:~$
```